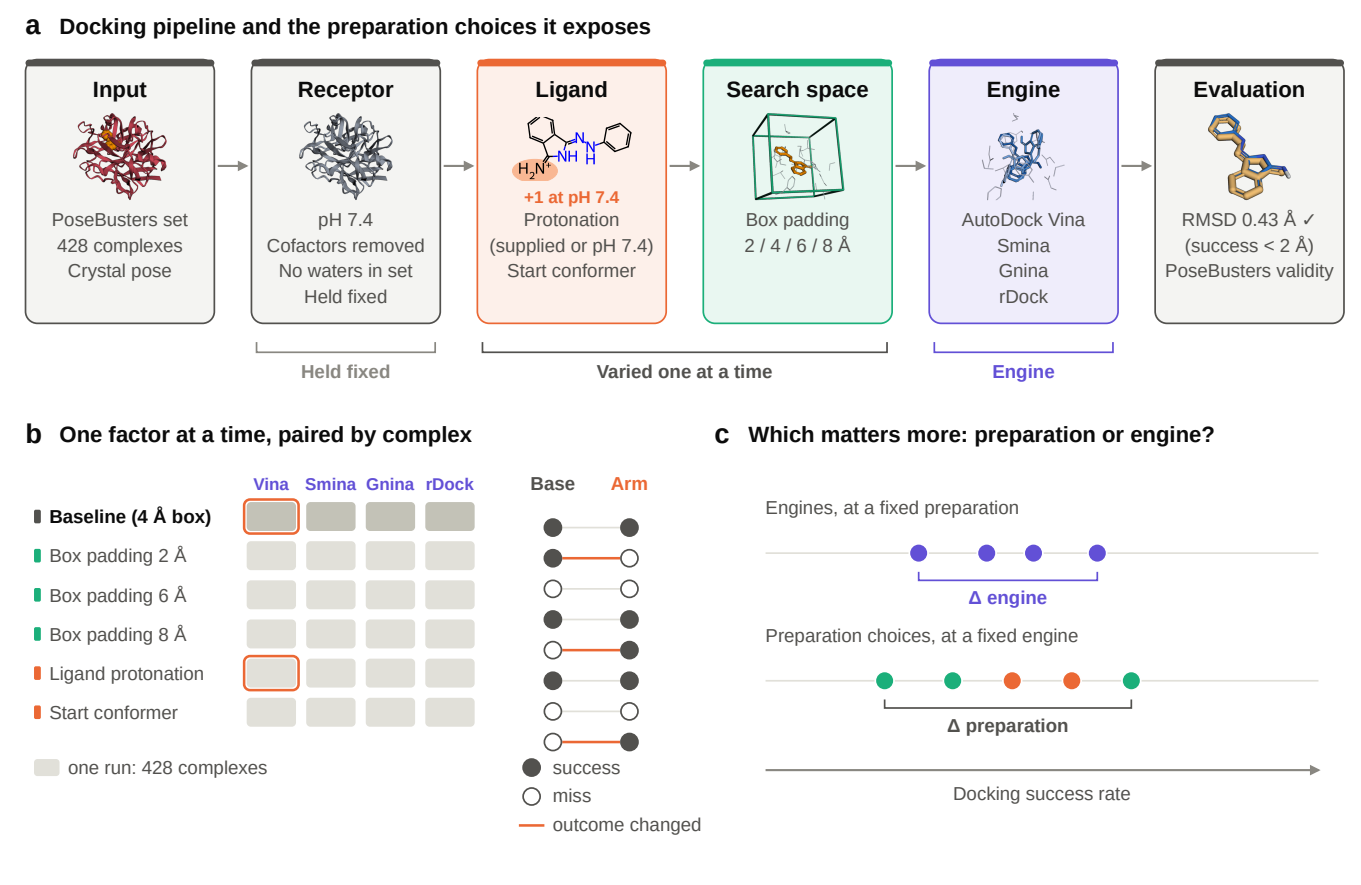

a Docking pipeline and the preparation choices it exposes
Input
PoseBusters set
428 complexes
Crystal pose
Receptor
pH 7.4
Cofactors removed
No waters in set
Held fixed
Ligand
+1 at pH 7.4
Protonation
(supplied or pH 7.4)
Start conformer
Search space
Box padding
2 / 4 / 6 / 8 Å
Engine
AutoDock Vina
Smina
Gnina
rDock
Evaluation
RMSD 0.43 Å ✓
(success < 2 Å)
PoseBusters validity
Held fixed
Varied one at a time
Engine
b One factor at a time, paired by complex
Vina
Smina
Gnina
rDock
Baseline (4 Å box)
Box padding 2 Å
Box padding 6 Å
Box padding 8 Å
Ligand protonation
Start conformer
one run: 428 complexes
Base
Arm
success
miss
outcome changed
c Which matters more: preparation or engine?
Engines, at a fixed preparation
Δ engine
Preparation choices, at a fixed engine
Δ preparation
Docking success rate